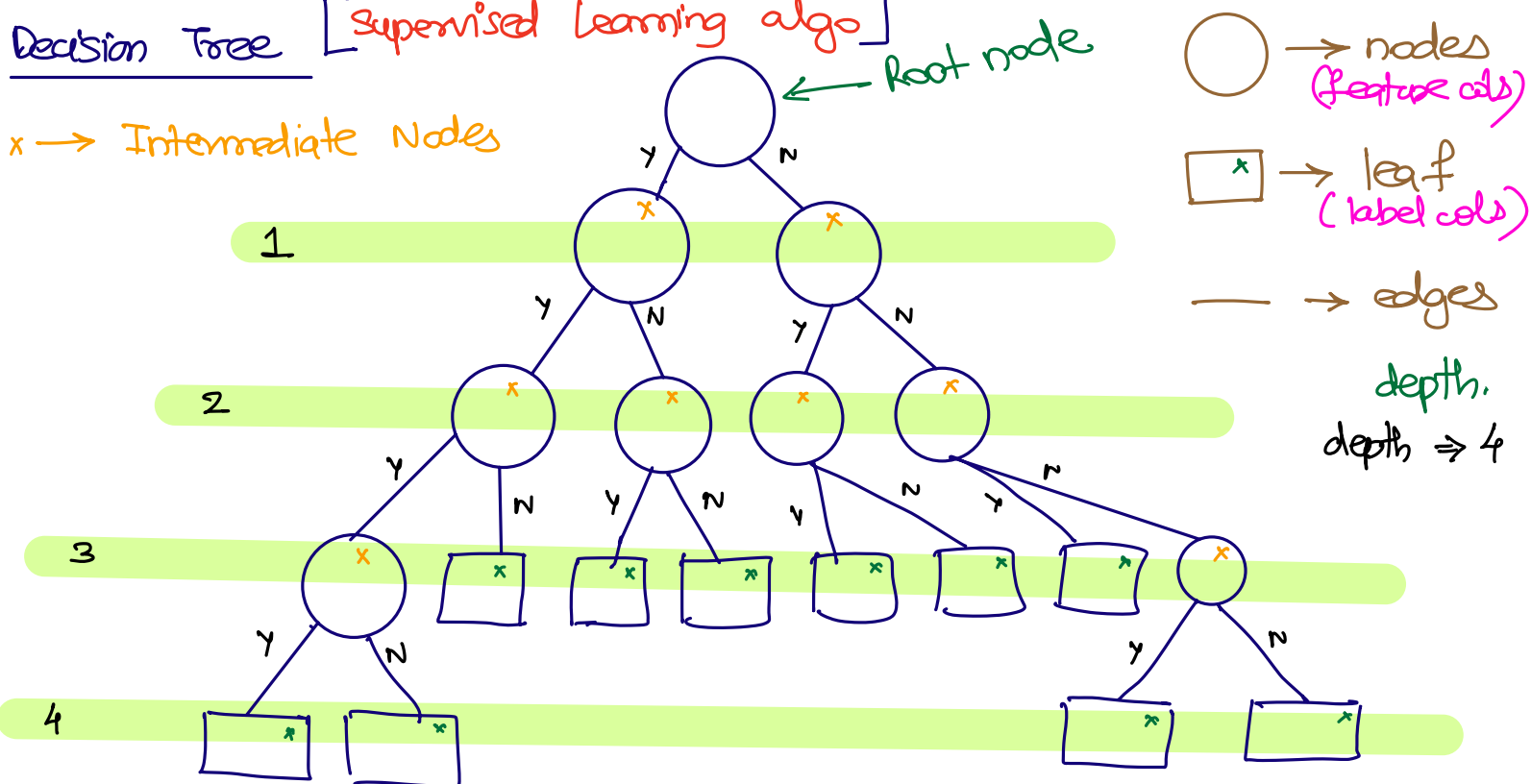
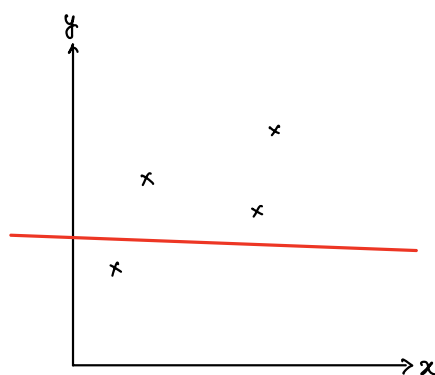


Decision Tree [Supervised Learning algo]

x → Intermediate Nodes



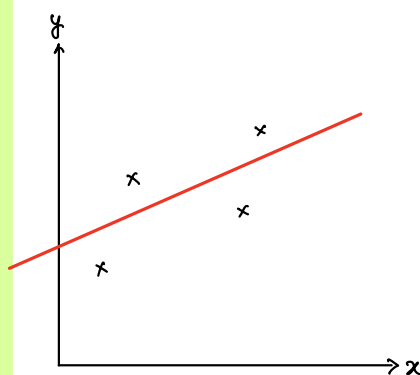
Pre-req



model 1

Underfitted Model

Any model that doesn't fit the CL criteria



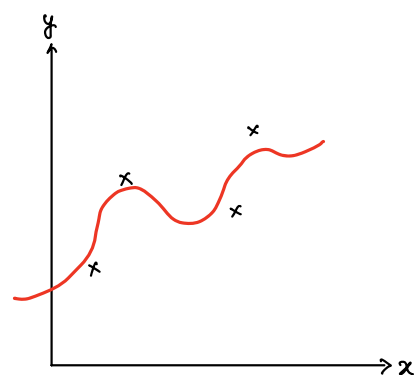
model 2

Generalized Model

—oo—

Balanced Model

model that understands population's pattern.



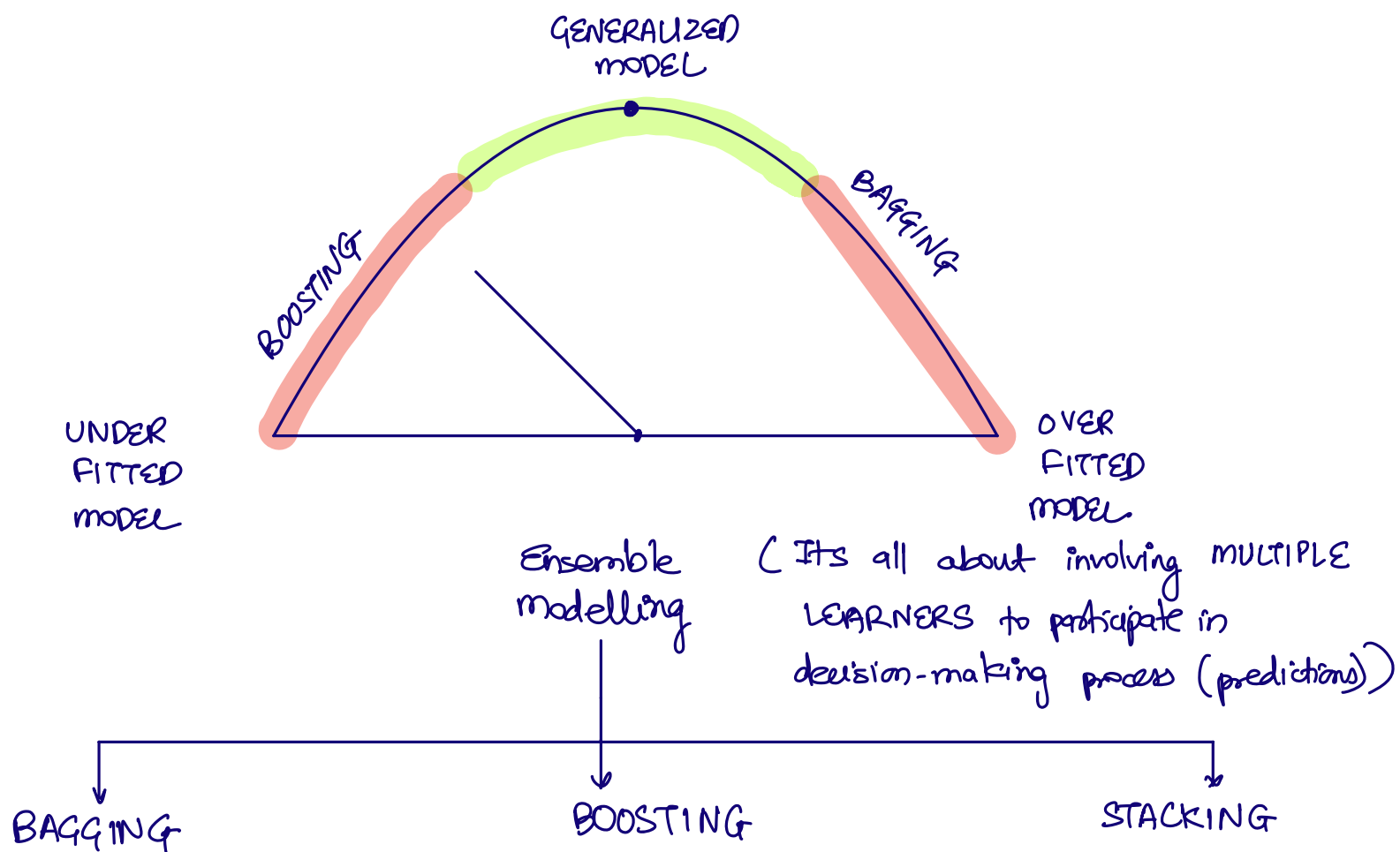
model 3

Overfitted model

model that kind of memorize the training data pattern.

Ensemble Modelling Techniques

These models ideally help derive **GENERALIZED** models.



Concept of Bagging

- **Bootstrap AGGregation**
- used to deal with **OVERFITTING** Problem
- if you perform **BAGGING**, there's a **CHANCE** of DERIVING a **GENERALIZED** model.

Bagging algo

Two hyperparameters:

① How much learners you want? $\longrightarrow 5$

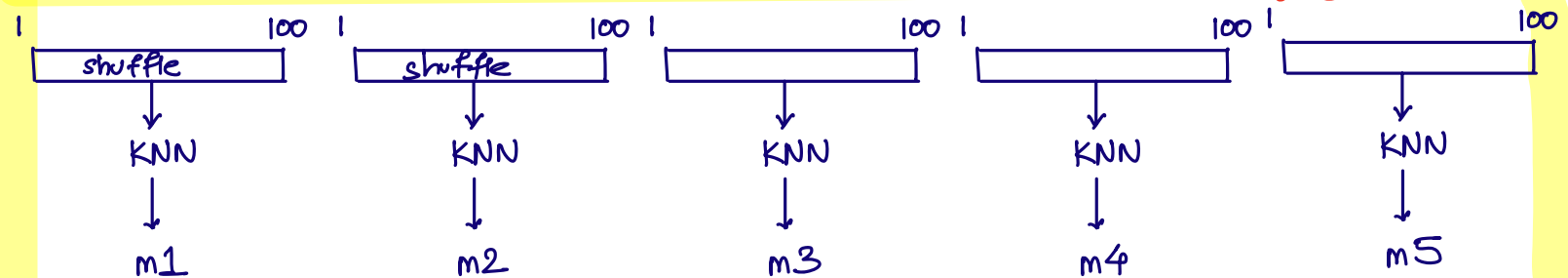
② Which ml algo to use to train each learner? $\longrightarrow$ KNN

eg: training dataset containing 100 records with categorical labels (h, s)

TRAINING PHASE

Shuffle method

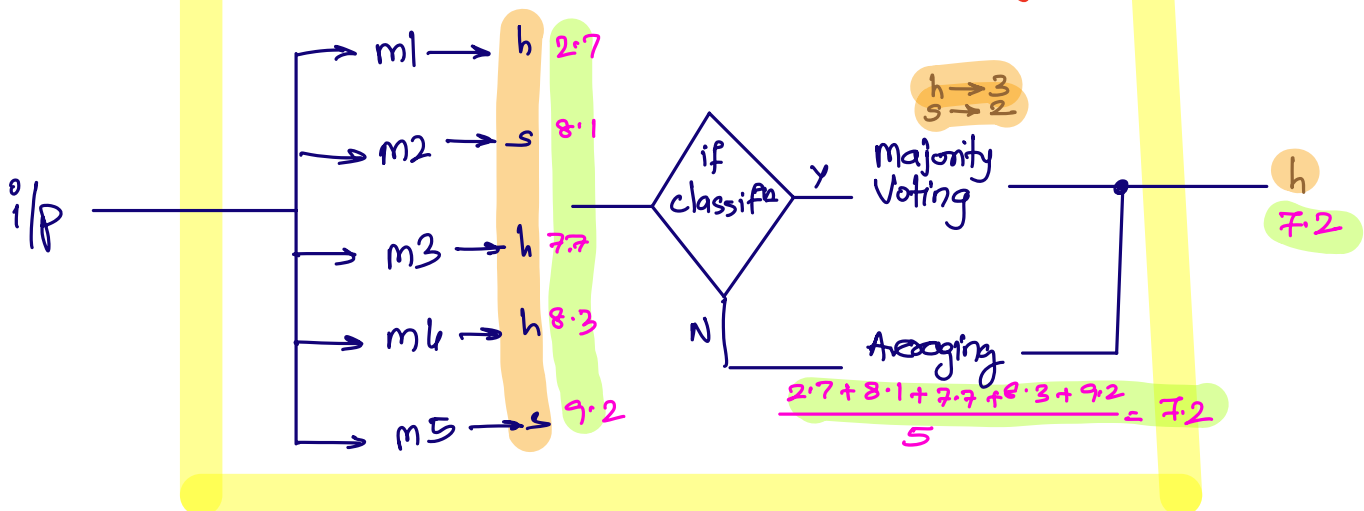
modelBagging



Prediction Phase

modelBagging

Regression
classification



Training in Bagging Creating learners.

SHUFFLE METHOD

- entire training set is considered for training individual learners.
- it uses RANDOM SHUFFLING to create each learner.

SAMPLING METHOD

create subset of the TRAINING SET for each learner.

Sampling with Replacement

Here the generated sample may have REPEATED RECORDS

eg: [1, 2, 3, 4, 5]

s1 → [1, 1, 4, 2]

s2 → [5, 1, 4, 2]

s3 → [1, 1, 1, 1]

Sampling without Replacement

Here the generated sample will have UNIQUE records only.

eg: [1, 2, 3, 4, 5]

s1 → [1, 5, 4, 2]

s2 → [5, 1, 4, 2]

s3 → [1, 2, 4, 3]